

Verification and Validation Report: Physiotherapy Companion

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1 Revision History

Name(s)	Date	Notes
Vaisnavi S.	March 9th, 2026	Created initial version of VnV Report
Vaisnavi S.	April 5th, 2026	Applied feedback to the VnV Report
Vaisnavi S.	April 6th, 2026	Finalized the final version of the VnV Report

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2 Symbols, Abbreviations and Acronyms

symbol	description
TC	Test Case
UT	Usability Test
PR	Performance Test
RFT	Reliability and Fault-Tolerance Test
SR	Security Test
MS	Maintainability and Support Test
FR	Functional Requirement
SR-AR	Security Access Requirement

3 Introduction

This document contains information about all the tests that were conducted to inspect PhysioCompanion's behaviour and output. There is some deviation from the original VnV plan, and the justifications are elaborated on within those sections. Functional tests are described initially, and those are followed by individual module or functionality unit tests. Changes that have been or are in the process of being implemented are followed after. The document ends with some code coverage statistics and their discussion.

4 Functional Requirements Evaluation

1. test-TC1 : Physiotherapist Registration (Valid License)

Initial State: App launched; registration screen accessible; Firebase Auth and Firestore active.

Input: `registerPhysio()` called with a valid license format and an active license in `validLicenses`.

Output: Auth user created. Firestore `users/{uid}` written with `role="physio"`, `verified=true`, and `createdAt`. All assertions passed.

Result: Pass

2. test-TC2 : Physiotherapist Registration (Invalid License Format)

Initial State: App launched; registration screen accessible.

Input: `registerPhysio()` called with an invalid license format (missing dash or incorrect digits).

Output: Error "Invalid license format" thrown. No Auth user or Firestore writes created.

Result: Pass

3. test-TC3 : Physiotherapist Registration (Unverified License)

Initial State: App launched; registration screen accessible.

Input: `registerPhysio()` called with valid format but license not found or `active=false` in `validLicenses`.

Output: Error “License not verified” thrown. No Auth user or Firestore writes created.

Result: Pass

4. **test-TC4 : Patient Registration (Valid Invite Code)**

Initial State: App launched; `inviteCodes` collection seeded with an active, unused code containing a `physioId`.

Input: `registerPatient()` called with a valid invite code.

Output: Auth user created. Firestore `users/{uid}` written with `role="patient"` and stored `physioId`. Invite code updated with `used=true` and `usedBy={uid}`.

Result: Pass

5. **test-TC5 : Patient Registration (Non-Existent Invite Code)**

Initial State: App launched; `inviteCodes` collection does not contain the submitted code.

Input: `registerPatient()` called with a code that does not exist in `inviteCodes`.

Output: Error “Invalid invite code” thrown. No Auth user or Firestore writes created.

Result: Pass

6. **test-TC6 : Patient Registration (Inactive Invite Code)**

Initial State: App launched; `inviteCodes` contains a matching code with `active=false`.

Input: `registerPatient()` called with an inactive invite code.

Output: Error “Invite code inactive” thrown. No Auth user or Firestore writes created.

Result: Pass

7. **test-TC7 : Patient Registration (Already Used Invite Code)**

Initial State: App launched; `inviteCodes` contains a matching code with `used=true`.

Input: `registerPatient()` called with an already used invite code.

Output: Error “Invite code already used” thrown. No Auth user or Firestore writes created.

Result: Pass

8. **test-TC8 : Patient Registration (Invite Code Missing Physio Link)**

Initial State: App launched; `inviteCodes` contains a matching code with no `physioId` field.

Input: `registerPatient()` called with a code missing a `physioId`.

Output: Error “Invite code missing physio link” thrown. No Auth user or Firestore writes created.

Result: Pass

9. **test-TC9 : User Login (Valid Physiotherapist Credentials)**

Initial State: Login screen available; Firestore contains a profile with `role="physio"`.

Input: `loginUser()` called with valid PT credentials.

Output: Login returns object with `uid` and profile data including `role="physio"`. No errors thrown.

Result: Pass

10. **test-TC10 : User Login (Valid Patient Credentials)**

Initial State: Login screen available; Firestore contains a profile with `role="patient"`.

Input: `loginUser()` called with valid patient credentials.

Output: Login returns object with `uid` and profile data including `role="patient"`. No errors thrown.

Result: Pass

11. **test-TC11 : User Login (Missing Firestore Profile)**

Initial State: Login screen available; valid Firebase Auth credentials exist but no Firestore profile.

Input: `loginUser()` called with valid credentials.

Output: Error “User profile missing in Firestore” thrown. Login rejected.

Result: Pass

12. **test-TC12 : User Logout**

Initial State: User session active.

Input: `logoutUser()` called.

Output: Auth sign-out called once and completes without error.

Result: Pass

13. **test-TC13 : Get User Data**

Initial State: Firestore populated with test user documents.

Input: `getUserData()` called with a valid uid, a non-existent uid, and an empty uid.

Output: Valid uid returns `UserData`. Non-existent uid returns null. Empty uid throws an error.

Result: Pass

14. **test-TC14 : User Lookup by Email**

Initial State: Firestore populated with test user documents.

Input: `getUserByEmail()` called with a registered email and an unregistered email.

Output: Registered email returns user object and true. Unregistered email returns null and false.

Result: Pass

15. **test-TC15 : Update User Data**

Initial State: Firestore populated with test user documents.

Input: `updateUser()` called with a valid update payload.

Output: Update returns true. Changes reflected in Firestore.

Result: Pass

16. **test-TC16 : Delete User**

Initial State: Firestore populated with test user documents.

Input: `deleteUser()` called with a valid uid, a valid email, and a non-existent case.

Output: Valid cases return true. Non-existent case returns false.

Result: Pass

17. **test-TC17 : User Queries**

Initial State: Firestore populated with linked PT and patient records.

Input: `getPatientsByPhysioId()`, `searchByName()`, and `getAllUsers()` called.

Output: Each function returns correctly filtered arrays. Errors return empty arrays.

Result: Pass

18. **test-TC18 : User Info Lookup**

Initial State: Firestore populated with linked PT and patient records.

Input: `getUserdbInfo()` called with name only and with name and birthday filter.

Output: Returns categorized patients and physiotherapists correctly. Birthday filter works as expected.

Result: Pass

19. **test-TC19 : PT Account Delete**

Initial State: Firestore populated with a valid physiotherapist and linked patient.

Input: `deletePTAccount()` called with a valid physio, a non-physio user, and a non-existent patient.

Output: Valid case deletes patient successfully. Non-physio and non-existent cases throw appropriate errors.

Result: Pass

20. **test-TC20 : User Authentication Validation**

Initial State: Firestore populated with test user credentials.

Input: `usernamePwMatch()` and `authenticateUser()` called with valid credentials, an invalid password, and a non-existent user.

Output: Valid credentials return true and user object. Invalid password throws an error. Non-existent user returns null.

Result: Pass

21. **test-TC21 : System Initialization**

Initial State: System not yet initialized.

Input: `initializeSystem()` called.

Output: Success message logged to console. No errors thrown.

Result: Pass

22. **test-TC22 : Credential Validation**

Initial State: User account exists in Firestore.

Input: `validateCredentials()` called with a correct password and an incorrect password.

Output: Correct password returns true. Incorrect password returns false.

Result: Pass

5 Nonfunctional Requirements Evaluation

5.1 Usability

1. **UT-1 : Navigation Clarity**

Initial State: App installed on a mobile device; user logged in.

Input: User was asked to find and open the exercise screen, start a recording session, and return to the dashboard without assistance.

Output: User successfully navigated through the required screens without confusion. Feedback indicated labels and buttons were clear.

Result: Pass

2. **UT-2 : Instruction Clarity During Exercise**

Initial State: User is on the record exercise screen with camera access enabled.

Input: User initiated an exercise session. The application displayed live on-screen feedback instructing the user of any adjustments.

Output: User reported that instructions were clear and prompts matched the required actions. Minor wording improvements were noted and updated.

Result: Pass

3. **UT-3 : Survey Questionnaire – ui_testing**

Initial State: App accessible to testers on mobile devices.

Input: Five users completed a short usability questionnaire addressing ease of use, clarity and navigation.

Output: Average rating across categories was $\geq 4/5$. Minor UI refinements (font size and button spacing) were applied based on feedback.

Result: Pass

4. **test-UT-4 : Account Creation (Usability)**

Initial State: App installed on participant's device; invite code and PT license number provided.

Input: Participant launched the app, selected their role, and completed account creation.

Output: All participants created accounts successfully without assistance. PT license and invite code fields completed without difficulty. Minor font size feedback noted and resolved.

Result: Pass

5. **test-UT-5 : Exercise Recording Initiation**

Initial State: Patient logged in on mobile device; camera permission granted; backend server running.

Input: Participant navigated to the Record screen and pressed Start Recording.

Output: Camera activated and pose detection began. On-screen prompts instructed participant to enter the detectable frame area. Navigation and button labels reported as clear.

Result: Pass

6. **test-UT-6 : Real-Time Pose Detection and Form Feedback**

Initial State: Patient positioned within camera frame; pose detection active; backend returning joint angle and ROM metrics.

Input: Participant performed knee extension repetitions.

Output: App detected relevant body landmarks and displayed appropriate form feedback overlays. Rep count incremented correctly. Detection accuracy was reported as satisfactory.

Result: Pass

7. **test-UT-7 : Exercise Page (Instructions)**

Initial State: Patient logged in; Exercises tab accessible.

Input: Participant navigated to the Exercises page and reviewed listed exercises.

Output: Exercises displayed with names, target muscle groups, set/rep counts, and step-by-step instructions. Instructions reported as clear and easy to follow.

Result: Pass

8. **test-UT-8 : Progress Screen (Session History)**

Initial State: Patient logged in with at least one completed session stored in Firestore.

Input: Participant navigated to the Progress tab.

Output: Session history displayed with ROM and rep count data. Layout was readable and logically arranged. Participants suggested adding sets and pain level as future enhancements.

Result: Pass

9. test-UT-9 : PT Dashboard (Patient List and Activity)

Initial State: Physiotherapist logged in; at least one linked patient with recorded sessions.

Input: PT participant reviewed the home dashboard and selected a patient to view their activity history.

Output: Dashboard displayed linked patients and activity history. Patient detail view showed session history and ROM chart. Layout reported as clear and sufficient for clinical use.

Result: Pass

10. test-UT-10 : PT Feedback Submission

Initial State: Physiotherapist logged in; patient session detail page accessible.

Input: PT participant selected a session entry and submitted a written comment.

Output: Feedback saved to Firestore and associated with the session. Comment visible from the patient's session view. PT participants rated the feature as clinically useful.

Result: Pass

5.2 Performance

1. PR-1 : Exercise Interface Response Time – ex_int_speed

Initial State: User logged in on a mobile device; backend server running.

Input: User started recording from the record screen. Screen recording timestamps were used to measure the time from tapping “Start” to the first visible feedback/result. There were five trials conducted.

Output: Measured response times were 1.87s, 1.92s, 1.79s, 1.95s, and 1.84s. The average end-to-end response time was 1.87 seconds (≤ 2 seconds).

Result: Pass

2. PR-2 : Repeated Use Performance

Initial State: Application running normally on mobile device.

Input: User completed 10 consecutive “Start → Record Exercise → Stop” cycles.

Output: There was no significant slowdown observed across the 10 trials. Additionally, the response times were all within acceptable range and no crashes took place.

Result: Pass

5.3 Reliability and Fault-Tolerance

1. RFT-1 : Backend Failure Handling – fail_recovery

Initial State: User actively using the application while connected to backend.

Input: Backend server was manually terminated during active analysis.

Output: Application displayed a network error message and remained responsive (no crash). After backend restart, the user was able to retry and continue normal operation.

Result: Pass

2. RFT-2 : Network Interruption During Upload – Interrupted_vid

Initial State: User uploading recorded exercise video.

Input: WiFi connection was disabled mid-upload and restored after approximately 10 seconds.

Output: Application detected upload failure and displayed a retry option. User retried successfully and uploaded video was verified to be intact.

Result: Pass

5.4 Security

1. SR-1 : Unauthorized Access to Protected Endpoint

Initial State: Backend server running with authentication enabled.

Input: A request was sent to a protected endpoint without authentication credentials.

Output: Server returned HTTP 401 Unauthorized. Additionally, no protected data was returned.

Result: Pass

2. SR-2 : Invalid Credentials Login

Initial State: Application login screen available.

Input: Login attempted using an email not registered in the database or incorrect password.

Output: System denied login and displayed an appropriate error message without exposing any system details.

Result: Pass

5.5 Maintainability and Support

1. MS-1 : Reproducible Setup

Initial State: Fresh machine with no prior installation.

Input: Project repository was cloned and installed using instructions provided in README (the frontend and backend setup).

Output: Application ran successfully without missing dependencies. All documented setup steps were sufficient to successfully reproduce the system.

Result: Pass

2. MS-2 : Code Review and Quality Assurance

Initial State: Revision 0 completed.

Input: Numerous peer code reviews were conducted through pull requests and issue tracking. The unit tests were executed to validate core functionality, including the metric calculations and functional logic. The UI components were reviewed for usability, clarity of instructions,

and error handling. Moreover, the identified issues were documented and resolved.

Output: The core functions passed unit testing. The UI refinements were implemented based on peer feedback. The code structure maintained clear separation between modules and frontend components. Overall, there were no major functional issues identified during review.

Result: Pass

6 Comparison to Existing Implementation

This section will not be appropriate for every project.

7 Unit Testing

The following section outlines the unit tests created for all of the modules.

7.1 Authentication

7.1.1 Account Creation Tests

The following tests in Table [1](#) verify the functionality of the account creation module for both physiotherapists and patients, including validation checks and successful user creation workflows.

ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC1	FR1.1, FR1.2	Register physiotherapist with valid license format and a license that exists in <code>validLicenses</code> with <code>active=true</code> .	Auth user is created. Firestore <code>users/{uid}</code> document is written with <code>role="physio"</code> , trimmed name, lowercased email, uppercased license number, <code>verified=true</code> , and <code>createdAt</code> .	All assertions pass.		Pass
TC2	FR1.1, FR1.2	Register physiotherapist with an invalid license format (e.g., a missing dash or incorrect digits).	Error "Invalid license format (example: ON-123456)." is thrown. No Auth user is created. No Firestore writes occur.	All assertions pass.		Pass
TC3	FR1.1, FR1.2	Register physiotherapist with valid format but license is not found in <code>validLicenses</code> or <code>active != true</code> .	Error "License not verified." is thrown. No Auth user is created. No Firestore writes occur.	All assertions pass.		Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC4	FR1.1, FR1.3, FR2, FR3	Register patient with invite code that exists in <code>inviteCodes</code> , is <code>active=true</code> , <code>used=false</code> , and includes a <code>physioId</code> .	Auth user is created. Firestore <code>users/{uid}</code> document is written with <code>role="patient"</code> , trimmed name, lowercased email, stored <code>physioId</code> , normalized invite code, and <code>createdAt</code> . Invite is updated with <code>used=true</code> and <code>usedBy={uid}</code> .	All assertions pass.		Pass
TC5	FR1.3, FR3	Register patient with an invite code that does not exist in <code>inviteCodes</code> .	Error " <code>Invalid invite code.</code> " is thrown. No Auth user is created. No Firestore writes/updates occur.	All assertions pass.		Pass
TC6	FR1.3, FR3	Register patient with an invite code that exists but <code>active=false</code> .	Error " <code>Invite code inactive.</code> " is thrown. No Auth user is created. No Firestore writes/updates occur.	All assertions pass.		Pass

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ID	Ref. Req.	Action	Expected Result	Actual result	Re-	Result
TC7	FR1.3, FR2	Register patient with an invite code that exists but <code>used=true</code> .	Error "Invite code already used." is thrown. No Auth user is created. No Firestore writes/updates occur.	All assertions pass.		Pass
TC8	FR1.3, FR2	Register patient with an invite code that exists but missing <code>physioId</code> .	Error "Invite code missing physio link." is thrown. No Auth user is created. No Firestore writes/updates occur.	All assertions pass.		Pass

Table 1: Account Creation Test Cases

7.1.2 Valid User Tests

The following tests in Table 2 verify the functionality of the valid user module.

ID	Ref. Req.	Action	Expected Result	Actual result	Re-	Result
TC9	FR1, FR1.1, SR-AR1	Login with valid credentials for a physiotherapist whose Firestore profile <code>users/{uid}</code> exists and contains <code>role="physio"</code> .	Login returns an object containing the <code>uid</code> and stored profile data (including role and user fields). No errors occur.	All assertions pass.		Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC10	FR1, FR1.1, SR-AR1	Login with valid credentials for a patient whose Firestore profile <code>users/{uid}</code> exists and contains <code>role="patient"</code> .	Login returns an object containing the <code>uid</code> and stored profile data (including role and user fields). No errors occur.	All assertions pass.		Pass
TC11	FR1, FR1.1, SR-AR1	Login with valid credentials where the Firestore profile <code>users/{uid}</code> does not exist.	Error <code>"User profile missing in Firestore."</code> is thrown and login is rejected.	All assertions pass.		Pass
TC12		Logout after a user session is active.	The Auth sign-out operation is called once and completes without error.	All assertions pass.		Pass

Table 2: Valid User Test Cases

7.2 User Account Tests

The tests below outline the User Account module functionality. Note that many new functions were added for this iteration. Currently, it serves as an interface between the database and the various other components. Additionally, there may be some overlap between the Account Creation module and Valid User module. While the modules themselves perform a few different functions, they are inextricably linked due to their handling of the same entity. For example, the Valid User module consists of tests the ensure successful Login and the User Account module test cases extend that to testing for invalid and non-existent users.

ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
USER MANAGEMENT FUNCTIONS						
TC13	FR23, FR21	Get User Data: Test with valid uid, non-existent uid, and empty uid.	Valid uid returns UserData; non-existent returns null; empty throws error.	All assertions pass.		Pass
TC14	FR21	User Lookup: Test get user by email and email existence check with registered and unregistered emails.	Registered email returns user and true; unregistered returns null and false.	All assertions pass.		Pass
TC15	FR2, FR6, FR22	Update User: Test update user data.	Valid update returns true.	All assertions pass.		Pass
TC16	FR4	Delete User: Test delete by uid and by email with valid and non-existent cases.	Valid cases return true; non-existent return false.	All assertions pass.		Pass
QUERY FUNCTIONS						
TC17	FR18	User Queries: Test get patients by physioId, search by name, and get all users.	Returns correctly filtered arrays; errors return empty arrays.	All assertions pass.		Pass
TC18	FR21, FR25	User Info Lookup: Test getUserdbInfo with name only and with name+birthday filter.	Returns categorized patients/physios correctly; filtering works as expected.	All assertions pass.		Pass
SPECIALIZED FUNCTIONS						
TC19	FR2, FR4, FR22	PT_Account_Delete: Test with valid physio, non-physio user, and non-existent patient.	Valid case deletes patient; non-physio throws error; non-existent throws error.	All assertions pass.		Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC20	SR-AR1, SR-PR1	Test usernamePwMatch and authenticateUser with valid credentials, invalid password, and non-existent user.	Valid returns true/user; throws appropriate errors; non-existent returns null.	All assertions pass.		Pass
TC21	SR-AR1, SR-AR2	System Initialization: Test initializeSystem.	Logs success message to console.	All assertions pass.		Pass
VALIDATION HELPERS						
TC22	FR1, SR-AR1	Test validateCredentials with correct and incorrect passwords.	Correct returns true; incorrect returns false.	All assertions pass.		Pass

Table 3: User Account Service Test Cases

7.3 Live Feedback Tests

The following tests in Table 4 verify the functionality of the live feedback module, including real-time corrective message generation, safety warning triggers, rep counting, and physiotherapist comment submission and retrieval.

ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC23	FR13, FR14	Submit joint angle measurements within the acceptable range for a knee extension rep.	No corrective feedback message is generated. Rep count increments by one. Positive reinforcement message returned.	All assertions pass.		Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC24	FR13, FR14	Submit joint angle measurements below the minimum threshold for knee extension (insufficient range of motion).	Corrective feedback message is generated (e.g., " Extend your knee further to complete the rep. "). Rep is not counted.	All assertions pass.		Pass
TC25	FR13, FR14	Submit joint angle measurements above the maximum safe threshold, simulating an overextension.	Corrective feedback message is generated (e.g., " Reduce the range of your movement. "). Rep is not counted.	All assertions pass.		Pass
TC26	FR16, PR-SCR1	Submit joint angle measurements that exceed the defined dangerous movement threshold.	Safety warning is triggered (e.g., " Stop immediately dangerous movement detected. "). Exercise session is halted.	All assertions pass.		Pass
TC27	FR14, PR-SL2	Submit a sequence of 5 consecutive valid reps and verify feedback generation latency for each.	Feedback is returned for each rep within 2 seconds. Rep count reflects 5 completed reps. No degradation in response time across reps.	All assertions pass.		Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC28	FR14	Invoke the feedback generator with no pose landmarks detected (user out of frame).	Feedback message returned is "Please step into the camera frame." No rep is counted. No safety warning triggered.	All assertions pass.		Pass
TC29	FR20, FR25	Physiotherapist submits a written comment on a completed patient session.	Comment is saved to Firestore and associated with the correct session document. No errors thrown.	All assertions pass.		Pass
TC30	FR19, FR25	Patient retrieves physiotherapist feedback for a session that has an associated PT comment.	Correct comment text is returned and associated with the matching session. Non-existent session returns null.	All assertions pass.		Pass

Table 4: Live Feedback Test Cases

7.4 Video Capture and Review Tests

The tests below outline the Video Capture and Review functionality. Some of the interactions with following tests as well as other modules may touch on the video capture and review module, but the tests below are focused on the core functionalities of the module itself.

ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC31	FR12	Call <code>get_pose_connections</code> (<code>legs_only=False</code>) and verify against <code>POSE_CONNECTIONS</code> .	<code>POSE_CONNECTIONS</code> and <code>LEG_CONNECTIONS</code> have length greater than 0. <code>LEG_CONNECTIONS</code> length is less than <code>POSE_CONNECTIONS</code> , and the set of connections from <code>get_pose_connections(False)</code> equals the set of <code>POSE_CONNECTIONS</code> .	All assertions pass.	Pass
TC32	FR12	Call <code>get_pose_connections</code> (<code>legs_only=True</code>) and verify against <code>LEG_CONNECTIONS</code> .	The connections returned equal the set of <code>LEG_CONNECTIONS</code> .	All assertions pass.	Pass
TC33	FR12	Call <code>extract_landmarks()</code> with a valid pose result containing 33 landmarks.	Returns a list of 33 landmarks. Each landmark has x, y, z values that are close to the expected i/33 values.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC34	FR26	Call <code>extract_landmarks()</code> with <code>None</code> and with an empty result (<code>pose_landmarks=None</code>).	Returns <code>None</code> for both.	All assertions pass.	Pass
TC35	FR7	Initialize <code>PoseCamera</code> with mocked <code>VideoCapture</code> and mocked <code>PoseLandmarker</code> .	<code>Camera</code> initializes successfully. <code>display_size</code> is set to (640, 480). <code>cap.set</code> is called to configure frame dimensions.	All assertions pass.	Pass
TC36	FR7	Call <code>get_frame()</code> with mocked camera that returns a numpy array frame.	Returns (True, frame) where frame is a numpy array. Frame is resized to display dimensions.	All assertions pass.	Pass
TC37	FR12	Call <code>process_pose()</code> with a mocked frame.	Returns a mock result. The pose detector's <code>detect</code> method is called once with the converted RGB image.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC38	FR14	Call <code>draw_landmarks()</code> with a mock result and a numpy array frame.	Returns the frame. OpenCV's <code>line</code> and <code>circle</code> functions are called to draw the pose landmarks.	All assertions pass.	Pass

Table 5: Video Capture and Review Test Cases

7.5 Computer Vision and Form Analysis Tests

The following tests in Table 6 verify the functionality of the computer vision and form analysis module by ensuring that joint angles are accurately calculated, knee angles are correctly extracted from pose landmarks, the analyzer calibrates to the user's range of motion, repetitions are counted properly, and summary metrics are generated as expected.

ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC39	FR12	Call <code>angle_deg()</code> with points forming a 90-degree angle: <code>a=(0,1)</code> , <code>b=(1,1)</code> , <code>c=(1,0)</code> .	Returns 90.0 degrees (within 0.01 tolerance).	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC40	FR12	Call <code>angle_deg()</code> with points forming a 45-degree angle: a=(0,1), b=(1,1), c=(0,0).	Returns 45.0 degrees (within 0.01 tolerance).	All assertions pass.	Pass
TC41	FR12	Call <code>angle_deg()</code> with points forming a 180-degree straight line: a=(0,1), b=(1,1), c=(2,1).	Returns 180.0 degrees (within 0.01 tolerance).	All assertions pass.	Pass
TC42	FR26	Call <code>angle_deg()</code> with None values for each parameter position: (None, (1,1), (1,0)), ((0,1), None, (1,0)), ((0,1), (1,1), None).	Returns None for all three cases.	All assertions pass.	Pass
TC43	FR12	Call <code>knee_angle_from_result()</code> with a valid mock result containing right leg landmarks (hip=24, knee=26, ankle=28) forming a 90-degree angle.	Returns a valid angle value between 0 and 180 degrees.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC44	FR12	Call <code>knee_angle_from_result()</code> with a valid mock result containing left leg landmarks (hip=23, knee=25, ankle=27) forming a 90-degree angle.	Returns a valid angle value between 0 and 180 degrees.	All assertions pass.	Pass
TC45	FR26	Call <code>knee_angle_from_result()</code> with None instead of a result object.	Returns None.	All assertions pass.	Pass
TC46	FR26	Call <code>knee_angle_from_result()</code> with a result object that has <code>pose_landmarks=None</code> .	Returns None.	All assertions pass.	Pass
TC47	FR26	Modify the mock result by setting the right knee landmark (index 26) to None, then call <code>knee_angle_from_result()</code> with <code>side="RIGHT"</code> .	Returns None due to missing landmark.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC48	FR26	Modify the mock result by setting the left knee landmark (index 25) to None, then call <code>knee_angle_from_result()</code> with <code>side="LEFT"</code> .	Returns None due to missing landmark.	All assertions pass.	Pass
TC49	FR26	Set the mock result's <code>pose_landmarks</code> to an empty list and call <code>knee_angle_from_result()</code> .	Returns None due to empty landmarks.	All assertions pass.	Pass
TC50	FR12	Initialize Analyzer with default values.	<code>started=False</code> , <code>min_angle=0.0</code> , <code>max_angle=0.0</code> , <code>is_calibrating=False</code> , <code>cal_start_ts=None</code> , <code>cal_duration_s=10.0</code> , <code>cal_min=None</code> , <code>cal_max=None</code> , <code>rep_count=0</code> , <code>current_rep=None</code> , <code>rep_durations=[]</code> , <code>rep_state=""</code> .	All assertions pass.	Pass
TC51	FR8, FR27	Call <code>start_calibration(duration_s=5.0)</code> .	<code>is_calibrating=True</code> , <code>cal_duration_s=5.0</code> , <code>cal_start_ts</code> is set, <code>cal_min=None</code> , <code>cal_max=None</code> .	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC52	FR8, FR27	During calibration, update with angles [30, 45, 60, 50, 70, 20].	<code>cal_min</code> updates to min value (20), <code>cal_max</code> updates to max value (70).	All assertions pass.	Pass
TC53	FR8, FR27	During calibration with duration 0.1s, update with angle 50 at t=100.05s (within duration), then angle 60 at t=100.16s (past duration).	First update: <code>is_calibrating</code> remains True. Second update: <code>is_calibrating</code> becomes False, <code>started</code> becomes True, <code>min_angle</code> =50, <code>max_angle</code> =60.	All assertions pass.	Pass
TC54	FR8, FR26	During calibration, <code>call_calibrate_ with_angle</code> (None).	Calibration ignores None values. <code>cal_min</code> and <code>cal_max</code> remain None.	All assertions pass.	Pass
TC55	FR8, FR16	Set <code>min_angle</code> =30.0, <code>max_angle</code> =70.0, <code>started</code> =True	<code>rep_state</code> is either "Extension" or empty.	All assertions pass.	Pass
TC56	FR8, FR14	With rep state "Extension", update with angle 35.	Rep state transitions to "Flexion". <code>current_rep</code> timestamp is updated.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC57	FR8, FR11	With rep state "Flexion" and <code>current_rep=100.0</code> , update with angle 68 at t=101.5s (1.5s later).	Rep state transitions to "Extension". Rep count increments to 1. Rep duration of 1.5s is recorded.	All assertions pass.	Pass
TC58	FR8, FR11	With rep state "Flexion" but <code>current_rep=None</code> , update with angle 68.	Rep state transitions to "Extension" but no rep is counted. Rep count remains 0.	All assertions pass.	Pass
TC59	FR8, FR12	Call <code>update(25)</code> and <code>update(75)</code> and verify min/max expansion via <code>_rep.update</code> .	<code>min_angle</code> updates to 25, <code>max_angle</code> updates to 75.	All assertions pass.	Pass
TC60	FR8, FR27	During calibration mode, call <code>update(45.0)</code> .	<code>_calibrate_with_angle</code> is called with 45.0.	All assertions pass.	Pass
TC61	FR8, FR12	Before <code>started</code> is set, call <code>update(45.0)</code> .	<code>min_angle</code> and <code>max_angle</code> are both set to 45.0. <code>started</code> becomes True.	All assertions pass.	Pass
TC62	FR8, FR27	With <code>started=True</code> , <code>min_angle=30.0</code> , <code>max_angle=70.0</code> , call <code>update(20.0)</code> .	<code>min_angle</code> updates to 20.0. <code>max_angle</code> remains 70.0.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC63	FR8, FR27	With <code>started=True</code> , <code>min_angle=30.0</code> , <code>max_angle=70.0</code> , call <code>update(80.0)</code> .	<code>max_angle</code> updates to 80.0. <code>min_angle</code> remains 30.0.	All assertions pass.	Pass
TC64	FR8, FR26	With <code>started=True</code> , call <code>update(None)</code> .	Values remain unchanged: <code>min_angle=30.0</code> , <code>max_angle=70.0</code> .	All assertions pass.	Pass
TC65	FR11, FR17	With <code>min_angle=30.0</code> , <code>max_angle=70.0</code> , <code>rep_count=5</code> , <code>rep_state="Extension"</code> , <code>rep_durations=[1.2, 1.3, 1.1, 1.4, 1.2]</code> , call <code>summary()</code> .	Summary contains all expected keys: <code>min_degree</code> , <code>max_degree</code> , <code>rom_degree</code> , <code>calibrating</code> , <code>cal_time_left</code> , <code>rep_count</code> , <code>current_rep_duration</code> , <code>avg_rep_duration</code> , <code>rep_state</code> .	All assertions pass.	Pass
TC66	FR11, FR17	With <code>min_angle=30.123</code> , <code>max_angle=70.456</code> , <code>rep_count=5</code> , <code>rep_state="Extension"</code> , <code>rep_durations=[1.2, 1.3, 1.1, 1.4, 1.2]</code> , call <code>summary()</code> .	<code>min_degree</code> rounds to 30.1, <code>max_degree</code> rounds to 70.5, <code>rom_degree</code> calculates to 40.3.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC67	FR8, FR27	During calibration with <code>cal_start_ts=100.0</code> , <code>cal_duration_s=10.0</code> , at current time 105.0, call <code>summary()</code> .	<code>calibrating=True</code> , <code>cal_time_left=5.0</code> .	All assertions pass.	Pass
TC68	FR11, FR17	When not calibrating, call <code>summary()</code> .	<code>calibrating=False</code> , <code>cal_time_left=0.0</code> .	All assertions pass.	Pass
TC69	FR11, FR17	With <code>rep_durations=[]</code> (no reps completed), call <code>summary()</code> .	<code>avg_rep_duration=0</code> , <code>current_rep_duration=0</code> .	All assertions pass.	Pass
TC70	FR11, FR8	With <code>current_rep=100.0</code> and <code>rep_durations=[1.2, 1.3]</code> , at current time 101.5, call <code>summary()</code> .	<code>current_rep_duration=1.5</code> , <code>avg_rep_duration=1.25</code> .	All assertions pass.	Pass
TC71	FR8, FR12	With <code>min_angle=50.0</code> , <code>max_angle=50.0</code> , <code>started=True</code> , call <code>_rep_update(50.0)</code> .	Function handles gracefully without division by zero errors.	All assertions pass.	Pass
TC72	FR12, FR26	With <code>started=True</code> , <code>min_angle=-10.0</code> , <code>max_angle=30.0</code> , update with angle -20.0 then 40.0.	<code>min_angle</code> expands to -20.0, <code>max_angle</code> expands to 40.0.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC73	FR12, FR26	With <code>started=True</code> , <code>min_angle=0.0</code> , <code>max_angle=180.0</code> , update with angle 200.0.	<code>max_angle</code> expands to 200.0.	All assertions pass.	Pass
TC74	FR8	With <code>min_angle=50.0</code> , <code>max_angle=50.0</code> , <code>started=True</code> , <code>rep_state="Extension"</code> , call <code>_rep_update(50.0)</code> .	No state change occurs. Rep state remains "Extension". Rep count remains 0.	All assertions pass.	Pass

Table 6: Computer Vision and Form Analysis Test Cases

7.6 Performance and Latency Tests

The following tests in Table 7 verify the functionality of the performance and latency module by ensuring that the backend services initialize correctly, API endpoints respond appropriately to valid and invalid requests, frame processing handles mirroring and landmark detection correctly, frontend services communicate with the backend, and exercise metrics are saved to Firestore with proper error handling.

ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC75	FR12	Start the PoseBackend with mocked values.	PoseBackend init values work as intended successfully and is_calibrating is set to True.	All assertions pass.	Pass
TC76	FR8	Call reset() method on PoseBackend.	Previous analyzer is replaced with a newer analyzer object. New one has is_calibrating set to True.	All assertions pass.	Pass
TC77	FR8	Send POST request to /reset endpoint.	Response status 200 with JSON {"ok": true} . Backend reset method is called.	All assertions pass.	Pass
TC78	FR7, FR26	Send POST request to /process_frame with empty image data.	Response status 400 with error message "Missing imageBase64" . Landmarks is null.	All assertions pass.	Pass
TC79	FR7, FR26	Send POST request to /process_frame with invalid base64 image string.	Response status 400 with error message. No frame processing occurs.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC80	FR7, FR12	Send POST request to <code>/process_frame</code> with valid base64 image, <code>side="RIGHT"</code> , <code>mirrored=true</code> , <code>legsOnly=true</code> . Landmarks detected.	Response status 200. Contains all landmarks (33). Metrics contain correct values (angle=45.6, rep_count=5, rep_state="Extension").	All assertions pass.	Pass
TC81	FR7, FR26	Send POST request to <code>/process_frame</code> with valid image but no pose landmarks detected.	Response status 200. Landmarks is null. Default metrics returned with <code>calibrating=True</code> .	All assertions pass.	Pass
TC82	FR7	Send POST request to <code>/process_frame</code> with <code>mirrored=True</code> and verify frame flipping.	Frame is flipped horizontally before processing. <code>cv2.flip</code> is called once with the frame.	All assertions pass.	Pass
TC83	FR7	Send POST request to <code>/process_frame</code> with <code>mirrored=False</code> and verify no frame flipping.	Frame is not flipped. <code>cv2.flip</code> is not called.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC84	FR7	Test <code>b64_to_bgr_image</code> helper function with a valid test image.	Function correctly decodes base64 string back to a numpy array with expected dimensions (100, 100, 3).	All assertions pass.	Pass
TC85	FR12	Send POST requests with <code>side="RIGHT"</code> and <code>side="LEFT"</code> parameters.	<code>knee_angle_from_result</code> is called twice with correct side parameters: first with 'RIGHT', then with 'LEFT'.	All assertions pass.	Pass
TC86	FR7, FR26	Send POST request with valid image but simulate decoding error by returning None.	Response status 400 with error message in JSON.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC87	FR7	Call <code>processFrame()</code> with front camera facing, RIGHT side, and a valid base64 image. Mock successful API response.	Function calls <code>axios.post</code> with correct URL, payload includes <code>mirrored: true</code> , <code>side: "RIGHT"</code> , <code>legsOnly: true</code> . Returns the mocked response data with landmarks, connections, and metrics.	All assertions pass.	Pass
TC88	FR7	Call <code>processFrame()</code> with back camera facing, LEFT side.	Function calls <code>axios.post</code> with <code>mirrored: false</code> in the payload.	All assertions pass.	Pass
TC89	FR7	Call <code>processFrame()</code> and simulate a network error.	Function throws an error with message "Network error".	All assertions pass.	Pass
TC90	FR8	Call <code>resetBackend()</code> and mock successful response.	Function calls <code>axios.post</code> to <code>/reset</code> endpoint with timeout 8000ms. Returns successfully.	All assertions pass.	Pass
TC91	FR8	Call <code>resetBackend()</code> and simulate a failure.	Function throws an error with message "Reset failed".	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC92	FR1, FR1.1	Call <code>saveMetrics()</code> when no user is logged in (<code>getAuth</code> returns null).	Function throws error "User must be logged in to save metrics".	All assertions pass.	Pass
TC93	FR1.1, FR21	Call <code>saveMetrics()</code> when user is logged in but user document does not exist in Firestore.	Function throws error "User data not found in Firestore".	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC94	FR21, FR11	Call <code>saveMetrics()</code> with valid metrics data and mocked Firebase responses.	Returns document ID "mock-doc-id". Verifies Firestore calls: doc, getDoc, collection, addDoc. Saved data contains correct fields including actid with timestamp, analysis="pending", exercise="Knee Extension", completed_reps=8, max_height=70.5, min_height=30.1, duration=1, user info, current_angle=45.6, rom_degree=40.4, rep_state="Extension", avg_rep_duration=1.2, current_rep_duration=0.5.	All assertions pass.	Pass
TC95	FR11, FR21	Call <code>saveMetrics()</code> with minimal metrics data (missing optional avg_rep_duration and current_rep_duration).	Saved data has avg_rep_duration=0, current_rep_duration=0, duration=0.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC96	FR1.1, FR21	Call <code>saveMetrics()</code> when user document exists but has no name field.	Saved data uses "Unknown User" for the name field.	All assertions pass.	Pass
TC97	FR1	Call <code>saveSessionData()</code> when no user is logged in.	Function throws error "User must be logged in".	All assertions pass.	Pass
TC98	FR21, FR17, FR11	Call <code>saveSessionData()</code> with valid session data.	Returns document ID "mock-doc-id". Saved data contains actid with "session_" prefix, analysis="completed", completed_reps=8, duration=1, exercise="Knee Extension Session", max_height=70.5, min_height=30.1, average_rom=40.4, total_reps=8, avg_rep_duration=1.2.	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC99	FR11, FR21	Call <code>saveSessionData()</code> without <code>avgRepDuration</code> (uses session duration instead).	Saved data has duration calculated from <code>startTime</code> and <code>endTime</code> (0 seconds), and <code>avg_rep_duration</code> is undefined.	All assertions pass.	Pass
TC100	FR1.1, FR21	Call <code>saveSessionData()</code> when user document does not exist.	Saved data uses "Unknown User" for the name field.	All assertions pass.	Pass
TC101	FR1.1, FR21	Call <code>getUserDisplayName()</code> with logged in user and existing user document with name field.	Returns "Test User". Verifies doc and <code>getDoc</code> calls with correct parameters.	All assertions pass.	Pass
TC102	FR1.1, FR21	Call <code>getUserDisplayName()</code> when user document exists but has no name field.	Returns user email "test@example.com".	All assertions pass.	Pass
TC103	FR1	Call <code>getUserDisplayName()</code> when no user is logged in.	Returns "Unknown User".	All assertions pass.	Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Result
TC104	FR1.1	Call <code>getUserDisplayName()</code> and simulate a Firestore error.	Returns "Unknown User" (error caught and handled gracefully).	All assertions pass.	Pass

Table 7: Performance and Latency Test Cases

7.7 Profile Activity Tests

The following tests in Table 8 verify the functionality of the profile activity module. This module is responsible for storing, retrieving, and handling user activity data, stored user data, summary data, and chart data. The tests ensure this functionality works as expected under various conditions, including valid inputs, edge cases, and error scenarios.

ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC105	FR21, FR23	Test <code>getCurrentUser</code> with correct inputs.	Returns the correct user object.	All Assertions Pass		Pass
TC106	FR21, FR23	Test <code>getCurrentUser</code> with no current user.	Returns null.	All Assertions Pass		Pass
TC107	FR21, FR23	Test <code>getCurrentUser</code> with non-existent user.	Returns null.	All Assertions Pass		Pass
TC108	FR21, FR23	Test <code>getCurrentUser</code> with Firestore error.	Returns null.	All Assertions Pass		Pass
TC109	FR21, FR23	Test <code>getCurrentUserID</code> with correct inputs.	Returns the correct user ID.	All Assertions Pass		Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC110	FR21, FR23	Test getCurrentUserID with no current user.	Returns null.	All Assertions Pass		Pass
TC111	FR10, FR11, FR17, FR18, FR21	Test getUserActivities with correct inputs.	Returns the correct array of activities.	All Assertions Pass		Pass
TC112	FR10, FR11, FR17, FR18, FR21	Test getUserActivities with no activities.	Returns an empty array.	All Assertions Pass		Pass
TC113	FR10, FR11, FR17, FR18, FR21	Test getUserActivities with Firestore error.	Returns an empty array.	All Assertions Pass		Pass
TC114	FR10, FR11, FR17, FR18, FR21	Test getUserActivities with incorrect uid.	Returns an empty array.	All Assertions Pass		Pass
TC115	FR10, FR11, FR17	Test getUserSummary with correct inputs.	Returns the correct summary data.	All Assertions Pass		Pass
TC116	FR10, FR11, FR17	Test getUserSummary with correct inputs but duplicate dates.	Returns the correct summary data.	All Assertions Pass		Pass

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ID	Ref. Req.	Action	Expected Result	Actual sult	Re-	Result
TC117	FR10, FR11, FR17	Test getUserSummary with Firestore error.	Returns zero summary data.	All Assertions Pass		Pass
TC118	FR10, FR18	Test repsChartData with correct inputs.	Returns the correct chart data.	All Assertions Pass		Pass
TC119	FR10, FR18	Test repsChartData with correct inputs and duplicate dates.	Returns the correct chart data.	All Assertions Pass		Pass
TC120	FR10, FR18	Test repsChartData with activities but no reps.	Returns all 0 chart values.	All Assertions Pass		Pass
TC121	FR10, FR18	Test repsChartData with no activites.	Returns all 0 chart values.	All Assertions Pass		Pass
TC122	FR10, FR18	Test repsChartData with firestore error.	Returns all 0 chart values.	All Assertions Pass		Pass
TC123	FR10, FR18	Test exerciseChartData with correct data.	Returns the correct chart data.	All Assertions Pass		Pass
TC124	FR10, FR18	Test exerciseChartData with correct data and duplicate dates.	Returns the correct chart data.	All Assertions Pass		Pass
TC125	FR10, FR18	Test exerciseChartData with no activities.	Returns all 0 chart values.	All Assertions Pass		Pass
TC126	FR10, FR18	Test exerciseChartData with firestore error.	Returns all 0 chart values.	All Assertions Pass		Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC127	FR10, FR11, FR18, FR19, FR22	Test getComments with correct inputs and data.	Returns the correct comment data.	All Assertions Pass		Pass
TC128	FR10, FR11, FR18, FR19, FR22	Test getComments with no existing comments.	Returns an empty array.	All Assertions Pass		Pass
TC129	FR10, FR11, FR18, FR19, FR22	Test getComments with firestore error.	Returns an empty array.	All Assertions Pass		Pass
TC130	FR10, FR11, FR18, FR19, FR22	Test postComment with correct inputs and data.	Calls collection and update functions with expected inputs.	All Assertions Pass		Pass
TC131	FR10, FR11, FR18, FR19, FR22	Test postComment with an empty comment.	Should not call collection, addDoc, or setDoc.	All Assertions Pass		Pass
TC132	FR10, FR11, FR18, FR19, FR22	Test postComment with firestore error.	Should not call setDoc, but should proceed until addDoc errors.	All Assertions Pass		Pass

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ID	Ref. Req.	Action	Expected Result	Actual sult	Re-	Result
TC133	FR18, FR21	Test getName with correct inputs and data.	Returns the corresponding user's name.	All Assertions Pass		Pass
TC134	FR18, FR21	Test getName with an unknown user.	Returns 'Unknown User'.	All Assertions Pass		Pass
TC135	FR18, FR21	Test getName with a user with no name.	Returns 'Unnamed User'.	All Assertions Pass		Pass
TC136	FR18, FR21	Test getName with firestore error.	Returns 'Unknown User'.	All Assertions Pass		Pass
TC137	FR17, FR18, FR19, FR22	Test setSelectedUserID with a valid userID.	Calls AsyncStorage with expected inputs.	All Assertions Pass		Pass
TC138	FR17, FR18, FR19, FR22	Test setSelectedUserID with an empty userID.	Does not call AsyncStorage.	All Assertions Pass		Pass
TC139	FR17, FR18, FR19, FR22	Test getSelectedUserID when a userID is stored.	Returns the correct stored userID.	All Assertions Pass		Pass
TC140	FR17, FR18, FR19, FR22	Test getSelectedUserID when no userID is stored.	Returns null.	All Assertions Pass		Pass
TC141	FR17, FR18, FR19, FR22	Test getUser when a user is stored.	Returns the stored UserData structure.	All Assertions Pass		Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC142	FR17, FR18, FR19, FR22	Test <code>getSelectedUser</code> when no user is stored.	Returns null.	All Assertions Pass		Pass
TC143	FR17, FR18, FR19, FR22	Test <code>clearSelectedUserID</code> on AsyncStorage.	Calls <code>AsyncStorage.removeItem</code> with expected inputs.	All Assertions Pass		Pass
TC144	FR2, FR3	Test <code>getPhysioInviteCode</code> when an invite code exists.	Returns the corresponding physio invite code.	All Assertions Pass		Pass
TC145	FR2, FR3	Test <code>getPhysioInviteCode</code> when no invite code exists.	Returns null.	All Assertions Pass		Pass
TC146	FR2, FR3	Test <code>getPhysioInviteCode</code> when <code>userID</code> is invalid.	Returns null.	All Assertions Pass		Pass

Table 8: Profile Activity Test Cases

7.8 Exercise Data Tests

The following tests in Table 9 verify the functionality of the exercise data module. This module is responsible for storing, retrieving, and handling user exercise data, general exercise data for assignable exercises by physios, and assigned exercises for given users. This function also facilitates the modification, addition, and deletion of exercises for users by their assigned physios. The tests outlined in the table ensure that all of this functionality works as expected under various conditions, including valid inputs, edge cases, and error scenarios.

ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC147	FR5, FR18, FR21, FR27	Tests getExercisesById with valid userID and exercise data	Returns the corresponding user's exercises.	All Assertions Pass		Pass
TC148	FR5, FR18, FR21, FR27	Tests getExercisesById with valid userID but no exercise data	Returns undefined.	All Assertions Pass		Pass
TC149	FR5, FR18, FR21, FR27	Tests getExercisesById with firestore error	Returns undefined.	All Assertions Pass		Pass
TC150	FR5, FR18, FR21, FR27	Tests getExercise with valid exerciseID	Returns the corresponding exercise.	All Assertions Pass		Pass
TC151	FR5, FR18, FR21, FR27	Tests getExercise with an invalid exerciseID	Returns undefined.	All Assertions Pass		Pass
TC152	FR5, FR18, FR21, FR27	Tests getExercise with firestore error	Returns undefined.	All Assertions Pass		Pass
TC153	FR5, FR18, FR21, FR27	Tests getGeneralExercises when general exercises exist	Returns a list of all exercises.	All Assertions Pass		Pass
TC154	FR5, FR18, FR21, FR27	Tests getGeneralExercises when no general exercises exist	Returns undefined.	All Assertions Pass		Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC155	FR5, FR18, FR21, FR27	Tests getGeneralExercises with firestore error	Returns undefined.	All Assertions Pass		Pass
TC156	FR6, FR18, FR21	Tests removeUserExercise with valid userID and exercise data	Calls deleteDoc with the expected inputs.	All Assertions Pass		Pass
TC157	FR6, FR18, FR21	Tests removeUserExercise with valid userID but invalid exercise data	Does not call deleteDoc.	All Assertions Pass		Pass
TC158	FR6, FR18, FR21	Tests removeUserExercise with firestore error	Does not call deleteDoc.	All Assertions Pass		Pass
TC159	FR6, FR18, FR21	Tests addUserExercise with valid userID and exercise data	Calls addDoc and updateDoc with the expected inputs.	All Assertions Pass		Pass
TC160	FR6, FR18, FR21	Tests addUserExercise with firestore error	Does not call updateDoc.	All Assertions Pass		Pass
TC161	FR6, FR18, FR21	Tests updateUserExercise with valid userID and exercise data	Calls updateDoc with expected inputs.	All Assertions Pass		Pass
TC162	FR6, FR18, FR21	Tests updateUserExercise with valid userID but invalid exercise data	Does not call updateDoc with bad inputs.	All Assertions Pass		Pass

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ID	Ref. Req.	Action	Expected Result	Actual Result	Re-	Result
TC163	FR6, FR18, FR21	Tests updateUserExercise with firestore error	Does not call updateDoc.	All Assertions Pass		Pass
TC164	FR5, FR18, FR21, FR27	Tests getSelectedExercises with valid user ID and exercise data.	Returns corresponding exercise data.	All Assertions Pass		Pass
TC165	FR5, FR18, FR21, FR27	Tests getSelectedExercises with valid user ID but no exercise data.	Returns undefined.	All Assertions Pass		Pass
TC166	FR5, FR18, FR21, FR27	Tests getSelectedExercises with firestore error	Returns undefined.	All Assertions Pass		Pass

Table 9: Exercise Data Test Cases

8 Changes Due to Testing

There were a few changes that were implemented as a result of testing. Note that some of these changes are in progress.

8.1 Usability Testing Results

Analyses from usability testing revealed important user experience insights which has provided clarity with what should be the focus for further development. A subset of recommendations will be developed further, while others will be tabled as out of scope due to project timeline constraints. There were a total of 5 participants in the study, with 2 being PTs, and the rest were those who satisfied the pool criteria. The diverse perspectives proved to be essential by honing in on elements that may provide substantial added value to the application. Both user flows (PT and patient) were tested with the

corresponding participants.

The physiotherapist participants had 2 recommendations: A more comprehensive exercise activity record (eg. adding sets in addition to reps) for a clearer picture on the patient's performance, and inclusion of a post-exercise feedback form to provide an insight into performance. (eg. modifying the rep count due to patient's increased pain).

The other participants highlighted the need for audio cues while performing the exercise. Reading the instructions from the acceptable distance necessary for proper application function was troublesome. Secondly, additional data on the Progress tab would be beneficial for patients to keep track, in a non-technical context, of their recovery.

See [Usability Report](#) for additional information.

Note: The usability testing has been completed, but the document's Results and Discussion still remain.

9 Automated Testing

In terms of automated testing, all of the tests for the TypeScript frontend modules as well as the Python backend are automated. When it comes to running the tests, the Python tests are run using **Pytest** by simply typing **pytest** in the command line in the correct root project directory. These tests validate core backend functionality such as the metric calculations and helper functions utilized during exercise processing. Additionally, all of the TypeScript tests can be run in the same manner, however, they run with **Jest** and through running the following command: **npm run test** instead. Ultimately, these tests focus on validating frontend logic including authentication and user account management, user activity tracking, exercise data handling, performance statistics generation, and the feedback module that displays live feedback to users and allows both the physiotherapists and patients the opportunity to leave comments regarding the user's respective session. It is important to note that the both the Firebase calls and camera components are mocked during testing so that we are able to separately test the logic functionality. Also, the output results for both the frontend and backend automated test suites are printed to the console, ultimately allowing us to be able to quickly identify both the passing and failing tests immediately.

10 Trace to Requirements

The following section showcases the mapping between the various tests performed to the specific requirements that they validate, ultimately displaying the traceability to requirements.

Table 10: Functional Requirements and Corresponding Test Cases

Test Cases	Functional Requirement(s)
<i>Account Creation Tests</i>	
TC1-3	FR1.1, FR1.2
TC4-8	FR1.1, FR1.3, FR2, FR3
TC9-11	FR1, FR1.1
<i>User Account Tests</i>	
TC13-16	FR2, FR4, FR6, FR21, FR22, FR23
TC17-18	FR18, FR21, FR25
TC19-21	FR2, FR4, FR22, SR-AR1, SR-AR2, SR-PR1
TC22	FR1, SR-AR1
<i>Live Feedback Tests</i>	
TC23-25	FR13, FR14
TC26	FR16
TC27-28	FR14
TC29	FR20, FR25
TC30	FR19, FR25
<i>Video Capture and Review Tests</i>	
TC31-33, 37	FR12
TC34	FR26
TC35-36	FR7

TC38	FR14
<i>Computer Vision and Form Analysis Tests</i>	
TC39-41, 43, 44, 50	FR12
TC42, 45-49	FR26
TC51-53, 60, 62, 63, 67	FR8, FR27
TC54, 64	FR8, FR26
TC55	FR8, FR16
TC56	FR8, FR14
TC57, 58, 70	FR8, FR11
TC59, 61, 71	FR8, FR12
TC65, 66, 68, 69	FR11, FR17
TC72, 73	FR12, FR26
TC74	FR8
<i>Performance and Latency Tests</i>	
TC75, 85	FR12
TC76, 77, 90, 91	FR8
TC78, 79, 81, 86	FR7, FR26
TC80	FR7, FR12
TC82-84, 87-89	FR7
TC92	FR1, FR1.1
TC93, 96, 100-102	FR1.1, FR21
TC94, 95, 99	FR11, FR21
TC97, 103	FR1
TC98	FR11, FR17, FR21
TC104	FR1.1, FR21
<i>Profile Activity Tests</i>	
TC105-110	FR21, FR23
TC111-114	FR10, FR11, FR17, FR18, FR21

TC115-117	FR10, FR11, FR17
TC118-126	FR10, FR18
TC127-132	FR10, FR11, FR18, FR19, FR22
TC133-136	FR18, FR21
TC137-143	FR17, FR18, FR19, FR22
TC144-146	FR2, FR3
<i>Exercise Data Tests</i>	
TC147-155, 164-166	FR5, FR18, FR21, FR27
TC156-163	FR6, FR18, FR21

Table ?? showcases both the functional requirements and their respective test cases.

Table 11: Usability Tests and Corresponding Requirements

Test ID	Non-Functional Requirement
UT-1	LFR-AP1
UT-2	UHR-EOU2, UHR-LRN1
UT-3	UHR-EOU1, LFR-ST1
UT-4	UHR-EOU1, UHR-LRN1
UT-5	UHR-EOU2, UHR-AR1
UT-6	UHR-UPL1, UHR-AR1
UT-7	UHR-LRN1, UHR-EOU1
UT-8	UHR-EOU1, LFR-ST1
UT-9	UHR-EOU1, LFR-AP1
UT-10	UHR-EOU1, UHR-UPL1

Table 11 maps each of the Usability tests to their respective non-functional requirements.

Table 12: Performance Tests and Corresponding Requirements

Test ID	Non-Functional Requirement
PR-1	PR-SL1, PR-SL2
PR-2	PR-SL1

Table 12 maps each of the Performance tests to their respective non-functional requirements.

Table 13: Reliability & Fault-Tolerance Tests and Corresponding Requirements

Test ID	Non-Functional Requirement
RFT-1	PR-RFT1
RFT-2	PR-RFT2

Table 13 maps each of the Reliability and Fault-Tolerance tests to their respective non-functional requirements.

Table 14: Security Tests and Corresponding Requirements

Test ID	Non-Functional Requirement
SR-1	SR-AR1
SR-2	SR-AR1

Table 14 maps each of the Security tests to their respective non-functional requirements.

Table 15: Maintainability & Support Tests and Corresponding Requirements

Test ID	Non-Functional Requirement
MS-1	OER-WE1, OER-PR1

Table 15 maps each of the Maintainability and Support tests to their respective non-functional requirements.

11 Trace to Modules

The following section displays the traceability of each of the test cases to the specific modules they validated, which ultimately showcases the overall verification of our system.

Table 16: Tests for all of the Modules

Test ID	Module
TC1-8	M1 (Account Creation)
TC9-12	M2 (Valid User)
TC13-22	M3 (User Account)
TC22-30	M8 (Feedback), M11 (UI)
TC31-38	M6 (Analysis Control Flow), M10 (Draw)
TC39-74	M6 (Analysis Control Flow), M7 (Analyze), M9 (Metrics)
TC75-104	M6 (Analysis Control Flow), M11 (UI)
TC105-146	M5 (User Activity)
TC147-166	M4 (Exercise Data)
UT-1, UT-2, UT-3, UT-4, UT-5, UT-6, UT-7, UT-8, UT-9, UT- 10, PR-1, PR-2, RFT-1, RFT-2	M11 (UI)
SR-1, SR-2	M1 (Account Creation), M2 (Valid User)
MS-1, MS-2	M1 (Account Creation), M2 (Valid User), M11 (UI)

Table 16 maps each of the test cases to their respective modules.

12 Code Coverage Metrics

The backend code has a total coverage of 97%, with all modules having coverage above 90%. The analyze.py module has 100% coverage, while draw.py,

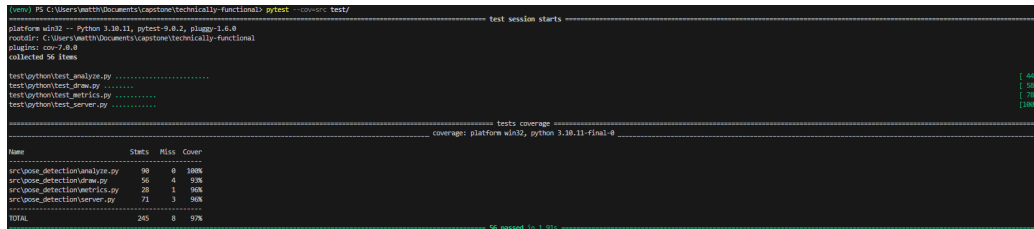


Figure 1: Python code coverage overview showing 97% total coverage across all modules.

Name	Stmts	Miss	Cover
src\pose_detection\analyze.py	90	0	100%
src\pose_detection\draw.py	56	4	93%
src\pose_detection\metrics.py	28	1	96%
src\pose_detection\server.py	71	3	96%
TOTAL	245	8	97%

Figure 2: Detailed Python code coverage by module showing analyze.py (100%), draw.py (93%), metrics.py (96%), and server.py (96%).

metrics.py, and server.py have 93%, 96%, and 96% coverage respectively. This indicates that the majority of the backend code is well-tested, with only a small portion of code not covered by tests.

Some areas of improvement would be to fully test each backend module to reach 100% coverage. This would be included in future testing plans but the current coverage points to the reliability of the backend code.

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	89.36	82.63	86.25	89.43	
authService.ts	97.72	100	80	97.43	32
exerciseData.ts	100	100	100	100	
firebase.ts	100	100	100	100	
metricsService.ts	97.91	78.04	100	97.72	157
poseService.ts	100	100	100	100	
profileActivity.ts	100	100	100	100	
roleStore.ts	0	0	0	0	1-18
temp.ts	0	0	0	0	1-35
useraccount.ts	83.69	80.48	81.25	83.6	66-67, 87-88, 94-95, 110-119, 175-176, 225-226, 288-289, 306-307, 334, 350-351, 363, 371, 390, 401, 435-436, 454, 464-465
Test Suites: 6 passed, 6 total					
Tests: 125 passed, 125 total					
Snapshots: 0 total					
Time: 4.015 s					
Ran all test suites.					

Figure 3: Detailed TypeScript Code Coverage.

The front end currently tested code has a total coverage of 80%, with major module implementations at or above 80% coverage. UserAccount module currently contains the lowest coverage as several of the functions were implemented originally and are now deprecated and unused, therefor not tested. The high coverage, at or above 97% for most of the modules, shows the testing's extensiveness into statement, branch, and function coverage.

There is room for improvement in the UserAccount module to remove the deprecated functions and increase the coverage. Additionally, there are some minor gaps in testing for some of the modules that could be filled in to reach even higher coverage. Overall, the current coverage indicates that the majority of the front end code is well-tested and will perform as intended.

File	% Stmts	% Branch	% Funcs	% Lines
All files	89.36	82.63	86.25	89.43
authService.ts	97.72	100	80	97.43
exerciseData.ts	100	100	100	100
firebase.ts	100	100	100	100
metricsService.ts	97.91	78.04	100	97.72
poseService.ts	100	100	100	100
profileActivity.ts	100	100	100	100
roleStore.ts	0	0	0	0
temp.ts	0	0	0	0
useraccount.ts	83.69	80.48	81.25	83.6

Test Suites: 6 passed, 6 total
 Tests: 125 passed, 125 total
 Snapshots: 0 total
 Time: 4.015 s
 Ran all test suites.

Figure 4: Detailed TypeScript Code Coverage Close Up.

13 Extras

The extras set out for the system consist of both a Design Thinking Document and a Usability Report. The Design Thinking Document aims to explain the process behind, and work done towards, the final design of the system. This document explains the overall flow of processing through the system and the interfacing elements, external libraries utilized, and more. The document flow is iterative, as is the design process, and aims to communicate the entire design process of the system. The Design Thinking Document can be found [here](#). The Usability Report aims to focus on evaluating the system from a user perspective. This report documents usability testing conducted with participants to assess ease of navigation, clarity of instructions, and overall

user experience. It includes observations, user feedback, and analysis of interaction patterns to identify the key areas to focus on and address. The findings are used to ensure the system meets the non-functional requirements and to guide improvements to the interface for both patients and physiotherapists. The Usability Report can be found [here](#).

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Reflection.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which parts of this document stemmed from speaking to your client(s) or a proxy (e.g. your peers)? Which ones were not, and why?
4. In what ways was the Verification and Validation (VnV) Plan different from the activities that were actually conducted for VnV? If there were differences, what changes required the modification in the plan? Why did these changes occur? Would you be able to anticipate these changes in future projects? If there weren't any differences, how was your team able to clearly predict a feasible amount of effort and the right tasks needed to build the evidence that demonstrates the required quality? (It is expected that most teams will have had to deviate from their original VnV Plan.)
5. Reflect on your use of CI for continuous integration testing (regression testing). Did your CI work to gatekeep your PRs? Would it have been helpful if you got your CI working earlier in the project? What are your plans to use CI for testing in future projects?

Vaisnavi:

- *What went well while writing this deliverable?*

One thing that went well while writing this deliverable was our ability to collaborate and efficiently split up the tasks and sections of this report. Specifically, in the case of the unit tests and traceability, we decided as a group for each member to derive the unit tests and traceability based on the sections of the app that they had worked on. This aided efficiency and productivity overall, and allowed the report to come together easily.

- *What pain points did you experience during this deliverable, and how did you resolve them?*

One of the biggest pain points I experienced during this deliverable was actually writing the unit tests. Writing the unit tests was actually harder than I had expected because mocking the Firebase database was something I had never done before. Since our authentication and valid user logic uses the Firebase database, we had to mock those database calls instead of using the real database, which took some time to learn and figure out. There was definitely a learning curve, but after looking through some examples and testing different approaches, I was able to get the mocks working correctly. All in all, this pain point allowed me to be able to understand how to test code that depends on external services.

Maham:

- *What went well while writing this deliverable?*

Our team worked really well together. Everyone completed their assigned tasks on time and were available to assist others if needed. We were all communicative, cordial and on schedule.

- *What pain points did you experience during this deliverable, and how did you resolve them?*

At the start, we split off into completing our 'Extras'. I was responsible for the Usability Testing and I had to ensure that there were results to work with. The team members conducted the tests on participants, while I worked on analysis as the results came in. Nearer to the deadline, we realized that the due date for Extras was moved to much later, (I don't think it was in the calendar until very recently), after which all the members hopped on deck to complete their corresponding module tests. Note to self: please double check your dates.

Eman:

- *What went well while writing this deliverable?*

One thing that went well while writing this deliverable was the team's ability to divide the work clearly based on ownership. Since each member had worked on specific parts of the application, it was natural to have each person derive the tests and evaluations for their respective sections. This made the process more efficient and ensured that the tests were written with a solid understanding of the underlying implementation. Communication within the team was consistent throughout, which helped us stay on track and address any blockers quickly.

- *What pain points did you experience during this deliverable, and how did you resolve them?*

One of the main pain points I experienced was setting up and debugging the network connection between the mobile device and the Flask backend during testing. The pose detection and recording functionality depended on the phone being able to reach the server, and resolving the IP configuration and network routing issues took considerable time. Once the connection was established, the full recording flow was tested with the real camera and backend, and the system performed as expected.

Matthew:

- *What went well while writing this deliverable?*

One thing that went well while working on this deliverable was our ability to communicate with each other and work together. We were able to make each meeting meaningful by updating the group on what we were working on, if we were stuck and also future deadlines. This allowed us to stay on track and stay on top of the deadlines. We also divided things based on our module implementation and that helped us clearly define the test case implementation and workload for the other documents.

- *What pain points did you experience during this deliverable, and how did you resolve them?*

A big pain point was working on the units test. I have not had much experience with testing and it has been a while since I wrote test cases for school. On top of that the difference in technology that we used academically to now was vastly different so I had to learn how to Mock certain objects and functionalities instead of using an actual database or object. I resolved them by just trying different things and looking at examples online. For example, I had to do a lot of reading on Mock and Patches for testing and the role that they play in testing. In the end I was able to create the unit test and learn a bit on how to test code.

Cieran:

- *What went well while writing this deliverable?*
Learning to use Jest and getting to put into practice all the things I learned from software testing was a great experience. We implemented several tests and explored sample test suite coverage in the class, but actually getting hands on and having to derive tests myself, imagine inputs, edge cases, error scenarios, and problematic inputs was a great learning experience. I think the entire team did a great job at writing tests for their respective modules and a great job on the report overall, completing it in a timely manner and communicating well throughout the process. Specifically, everyone came together for the usability testing to ensure the data we gathered came from as many unique sources and demographics as possible, and could lead to the design of many more implementable improvements both in and out of scope.

- *What pain points did you experience during this deliverable, and how did you resolve them?*

I was responsible for the traceability section of the document, one which required the input of my teammates to develop and came across some time management issues within my own limitations. Since the traceability section required the input of my teammates, I had to wait for them to complete their respective sections and derive their tests before I could complete the traceability. This was a bit of a bottleneck for me, but I was able to manage it by communicating with my teammates and ensuring that they were aware of the importance of completing their sections in a timely manner so that I could complete the traceability. Additionally, I had to manage my own time effectively to ensure that I could complete the traceability once I had all the necessary information from my teammates. Overall, it was a bit of a challenge, but through communication and time management, I was able to successfully complete the traceability section. Additionally, my own issue, I had very limited knowledge on Jest and TypeScript implementation and testing, so I had to learn how to use Jest on my own beginner implementations in TypeScript, which had a learning curve but I feel has taught me to make better TypeScript and React code overall.

Group

- *Which parts of this document stemmed from speaking to your client(s) or a proxy (e.g. your peers)? Which ones were not, and why?*

Parts of this document stemmed from speaking to our clients, specifically by receiving feedback from our peers who acted as proxies for physiotherapy patients. When demonstrating our application, they interacted with the system and gave relevant, useful feedback on the usability of the interface as well as how clear and easy to grasp the real-time exercise feedback was. This mainly had an impact on the sections related to usability testing, interface design, and how feedback is presented to users, especially the requirement for clear instructions and visible/audio cues during exercises. Moreover, the other parts of the document, such as functional requirement evaluation and technical implementation, did not come directly from speaking to our clients. These were based on the requirements defined in our SRS and the

testing done by our core team, since these parts relate more to the development logic and system reliability rather than user interaction.

- *In what ways was the Verification and Validation (VnV) Plan different from the activities that were actually conducted for VnV? If there were differences, what changes required the modification in the plan? Why did these changes occur? Would you be able to anticipate these changes in future projects? If there weren't any differences, how was your team able to clearly predict a feasible amount of effort and the right tasks needed to build the evidence that demonstrates the required quality? (It is expected that most teams will have had to deviate from their original VnV Plan.)*

Our VnV Plan deviated from the activities that were actually conducted in a few notable ways. The original plan outlined testing at a higher level of abstraction, targeting broader system functionalities rather than individual module functions. As development progressed and the codebase became more detailed, it became clear that more granular unit tests were necessary. This also coincided with our team further subdividing some of the modules during implementation, which resulted in more detailed tests being required. This led to the addition of the User Account module test suite, which did not exist in the original plan at all. As this module grew in scope and became central to both patient and PT workflows, dedicated testing for functions such as `getUserData()`, `updateUser()`, and `deletePTAccount()` became essential. The usability testing itself proceeded largely as planned, though the participant pool was smaller than initially intended due to scheduling constraints. The structured questionnaire format, pre-session, session, and post-session, was maintained as designed and provided consistent data across all five participants.

- *Reflect on your use of CI for continuous integration testing (regression testing). Did your CI work to gatekeep your PRs? Would it have been helpful if you got your CI working earlier in the project? What are your plans to use CI for testing in future projects?*

Our project currently uses GitHub Actions for CI, but as of right now, the workflow is limited and mainly displays the pull request activity and

build checks. With our implemented unit tests for both the frontend and backend, we want to integrate these directly into the CI pipeline. Specifically, the GitHub Actions workflow could be configured to automatically run the Jest test suite for the TypeScript frontend and the Pytest test suite for the Python backend whenever a pull request is opened or updated. In this case, if any test fails, the workflow would essentially fail, which would ultimately prevent changes from being merged until the issue is resolved. With this proposed change, this would make the CI pipeline much more effective for regression testing, as it would automatically verify that new code trying to be merged in does not break existing functionality. It is important to note that it would have been helpful to set this up earlier in the project, and as a result, for future projects, we would aim to do so. Thus, overall for future projects, we would plan to set up CI earlier and integrate our automated test suites (such as Jest and Pytest) so that the tests run automatically on every pull request and prevent merges if any tests fail.